Influence of Supersaturation on Particle size and Crystallite size of Hydroxyapatite Nanoparticles

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***Abstract*— In this study, we investigate the influence of supersaturation on the particle size and crystallite size of Hydroxyapatite nanoparticles, $[Ca_{10}(PO_4)_6(OH)_2]$synthesized using the wet chemical precipitation technique. For the synthesis process, different concentrations of calcium hydroxide and orthophosphoric acid were used to achieve different supersaturations. The Ca/P molar ratio was kept at 1.67 while changing the supersaturation. The synthesized nanoparticles were characterized using a laser particle analyzer, Scanning Electron Microscope (SEM), X-ray diffractometer (XRD), Energy Dispersive X-Ray Analysis (EDAX), and Fourier-Transform infrared (FT-IR) spectroscopy to study their size, morphology, structure, and composition. The results showed that the particle size and crystallite size of Hydroxyapatite nanoparticles can be controlled by varying the precursor concentrations to initiate different supersaturations.**

Keywords—Nano-Hydroxyapatite, wet chemical synthesis, supersaturation, Particle size, Crystallite size

I. Introduction

Hydroxyapatite (HA) nanoparticles have become increasingly popular in the field of biomedical engineering due to their unique physicochemical properties and potential applications in bone tissue engineering. The synthesis of HA nanoparticles has been widely investigated using various methods, including sol-gel, hydrothermal, and precipitation methods. The precipitation method is preferred among these methods due to its simplicity, cost-effectiveness, and ability to produce HA nanoparticles with morphology. This process can be easily modified and used in industrial applications. The precipitation reaction of hydroxyapatite in the wet chemical synthesis method is shown in (1). [1]

$$10\ Ca(OH)_2 + 6\ H_3PO_4 \rightarrow Ca_{10}(PO_4)_6(OH)_2 + 18\ H_2O \quad (1)$$

The biocompatibility of HA nanoparticles is an important factor to consider when designing biomaterials for bone tissue engineering. The biocompatibility of HA nanoparticles is influenced by several factors, including particle size, crystallite size, morphology, surface charge, and surface chemistry. Previous studies have substantiated that particle size plays a critical role in determining the biocompatibility of HA nanoparticles.[1] Smaller particles of HA have been shown to promote cell attachment, proliferation, and differentiation, making them a promising candidate for bone tissue engineering.[2]

A recent study based on classical nucleation theory has derived an equation to express the relationship between supersaturation and nucleation rate in nano HA synthesis by wet chemical synthesis technique. A major finding of the study was that the nucleation rate of a precipitating solution strongly depends on the supersaturation. And another finding of the study was final supersaturation is not increased with the increase of precursor concentrations of Calcium Hydroxide and Orthophosphoric acid.[3]

According to the finding of the study, it has been observed that there exists a direct relationship between supersaturation (S) and nucleation rate ($\frac{dN}{dt}$), as denoted by (2). As supersaturation increases the nucleation rate increases.

$$\frac{dN}{dt} = Aexp(-\frac{\beta v^2 \gamma^3}{(\kappa_B T)^3 (\ln|S|)^2}) \quad (2)$$

The present study is aimed to investigate the influence of supersaturation and nucleation rate on the crystallite size and the final particle size distribution of HA nanoparticles synthesized by the precipitation method. In this study, a series of five HA samples were synthesized with varying concentrations of calcium hydroxide and orthophosphoric acid, followed by the characterization of resulting the nanoparticles using a laser particle analyzer, Scanning Electron Microscopy (SEM), X-ray diffraction (XRD), and Fourier transform infrared spectroscopy (FT-IR). The results obtained from this study will contribute to understanding the influence of supersaturation on both the crystallite size and final particle size variation of HA nanoparticles. Furthermore, the findings will provide insight into the advancement of novel biomaterials intended for bone tissue engineering applications.

II. Materials and Methodology

A. Materials

H_3PO_4 Assay 88% (MERCK), Analytical grade $Ca(OH)_2$ Assay 98% (VWR), NH_4OH 28% (MERCK), De-ionized water as the solvent.

B. Research Methodology

The final particle size of nanoparticle synthesis by wet chemical synthesis process is controlled by successive nucleation and growth of HA. However, a recent study focused on the development of a mathematical equation with theories of classical nucleation theory for wet chemical synthesis of Hydroxyapatite nanoparticles reveals the influence of supersaturation on nucleation and final particle

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size. The study suggests supersaturated solutions provide a minimal effect on particle growth for the five HA samples prepared. The study provides evidence that the influence of growth, on final particle size is eliminated or minimal in achieving the precipitation under supersaturations in wet chemical synthesis.[3]

Therefore, this study was carried out to investigate the effect of nucleation rate which is controlled by the supersaturation on final particle size and crystallite size of HA.

C. Synthesis of n-HA

A suspension of Calcium Hydroxide 250 ml and a 250 ml Orthophosphoric acid solution were prepared using standard chemical reagents. The prepared Calcium Hydroxide suspension was subjected to stirring for one hour using a magnetic stirrer. Subsequently, the Orthophosphoric acid was added to the continuously stirring Calcium Hydroxide suspension at a controlled rate of 3ml/min. The pH variation of the suspension was recorded every 30 second intervals. After Orthophosphoric acid was added, the solution was sonicated for a duration of 30 minutes. Ammonium Hydroxide (NH_4OH) was introduced to the mixture to elevate the pH to 10 while continuous magnetic stirring. The pH was monitored while maintaining continuous stirring until a consistent pH value below 10 was attained, at which point the stirring was stopped, and the mixer was allowed to age for a period of 72 hours. [1], [3]

After a 72-hour aging period, the solvent layer above the precipitation was removed cautiously, and De-Ionized water was introduced to the precipitated beaker. The mixer was stirred for 1 min and particles were allowed to settle down. This process was repeated three times to neutralize the pH value of the slurry. The slurry was subjected to ultrasonic vibration for 30 minutes, followed by centrifugation at 5000 rpm for 6 minutes at a temperature of 30°C. Then the sample was oven dried at 60°C for a duration of 3 hours to facilitate drying. The oven-dried HA samples were subjected to calcinated at 600°C for 1 hour. As documented in Table 1, the five samples were prepared following the same method.[3]

Table 1. Precursors' concentrations in mol dm^{-3} for the prepared five HA samples

Sample / Reagent	1	2	3	4	5
$[Ca(OH)_2]$	0.167	0.334	0.501	0.630	0.840
$[H_3PO_4]$	0.100	0.200	0.300	0.371	0.494

III. Experimental

A. Laser Particle Analyzer

The wet HA specimen obtained from the precipitated slurry after sonication was subjected to particle size analysis using the HMK-CD2 Laser Particle Size Analyzer.

B. Fourier – Transformation Infrared Spectroscopy(FT-IR)

The dry HA specimen, obtained after oven–drying was subjected to analysis using the "Bruker Alpha II" compact FT–IR spectrometer. The analysis aimed to determine the functional groups can be found in synthesized nano HA in the 4000 – 400 cm-1 range using KBr as the carrier.

C. Scanning Electron Microscope

The morphology and size of the n-HA particles were investigated using a Scanning Electron Microscope (SEM) "Carl Zeiss evo 18". Prior to imaging, the HA samples were prepared by mounting them on stubs and gold sputter coated to provide the necessary conductivity for optimal SEM imaging.

D. Energy Dispersive X-Ray Analysis

The elemental compositions of Hydroxyapatite nanoparticles were analyzed through the application of Energy Dispersive X-ray Analysis (EDAX) with the use of AMETEK EDAX.

E. X-ray Diffraction Spectroscopy

X-ray diffraction (XRD) analysis was conducted on the oven-dried HA samples with the use of "Bruker D8 Advance" in 2θ angle range of 10° to 60°. Obtained data were used to calculate the crystallite size of each HA sample according to Scherrer's in (2).[4]

$$D = \frac{K\lambda}{\beta \cos\theta} \tag{2}$$

IV. Supersaturation

The highest supersaturation of solution is not determined by the highest precursor concentrations used. As the electrostatic interaction between ions of a saturated solution reduces the effective ionic concentration called "activity of the species" according to Debye-Huckel Limiting Law. Therefore, the following (3) were used to calculate the ionic strength of the solutions.[3]

$$\mathrm{I} = \frac{1}{2}\sum C_i z_i^2 \tag{3}$$

Where I, C_i and Z_i are the Ionic strength, concentration, and valence of the ions.
Activity for each species was calculated using the single ion molar ionic activity coefficient from (4). Where, γ_i is the single ion molar ionic activity coefficient.

$$\log\gamma_i = -0.5108 Z_i^2 \sqrt{I} \tag{4}$$

Activity product(Q) for Hydroxyapatite was calculated using the following (5) and (6) and it was used to calculate the supersaturation(S). Where, a is the ionic activity.

$$Ca_5(PO_4)_3(OH)_{(s)} \rightleftharpoons 5Ca^{2+}_{(aq)} + 3PO^{3-}_{4(aq)} + OH^-_{(aq)} \tag{5}$$

$$Q = [a_{Ca^{2+}}]^5 [a_{PO_4^{3+}}]^3 [a_{OH^-}] \tag{6}$$

Supersaturation for each sample was calculated as follows (7).

$$S = \frac{Q}{K_{sp}} \tag{7}$$

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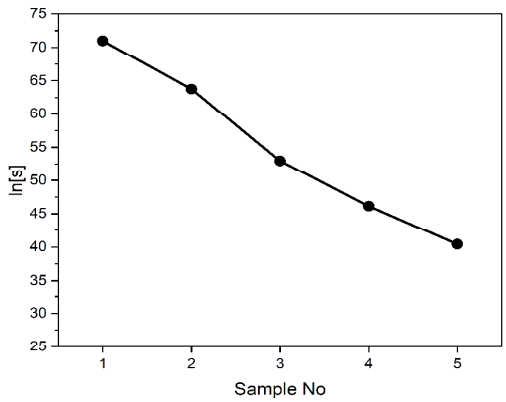

Fig. 1. Analytically calculated Supersaturation for five HA samples

Fig. 1. shows the supersaturation of each. According to the studies the highest supersaturation was calculated in the solution where the least precursor concentrations are used. This is due to that HA Sample 5 with highest precursor concentrations shows highest ionic strength and decrease the supersaturation.[3]

V. Results and Discussion

A. SEM Analysis

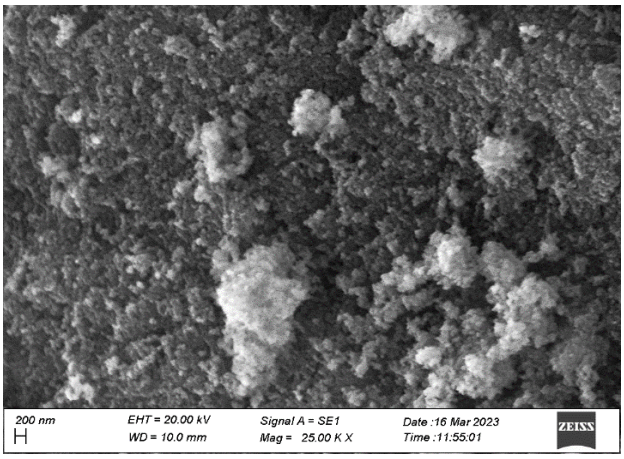

Fig. 2. SEM images of HA sample 1

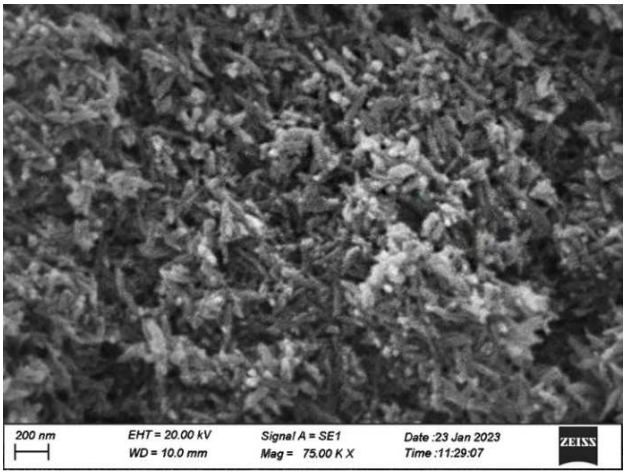

Fig. 3. SEM image of HA sample 3

The HA samples were gold sputtered before the Scanning Electron Microscopy analysis. The SEM analysis was performed.

It can be observed that the nanoparticles formed were highly agglomerated. Moreover, recent literature confirms that ultrasonic vibration on agglomerated particles in a slurry does not completely disperse into initially precipitated particles.[5]

Fig. 3. exhibits an enlarged, clear image that provides evidence of the formation of rod-shaped nanoparticles. Previous studies have indicated that there is a preference for rod shaped morphology in n-HA particles synthesized by wet chemical method.[6] These observations provide confirmation of n-HA formation through the wet chemical synthesis method. However, particle size variation assessment in each image is challenging due to the particles having a wide range of sizes and small deviations with respect to other synthesized samples. Furthermore, the magnified SEM images taken for samples exhibit significant blurriness and darker appearance. These issues can be due to the charging effects.

B. Fourier transform Infrared spectroscopy

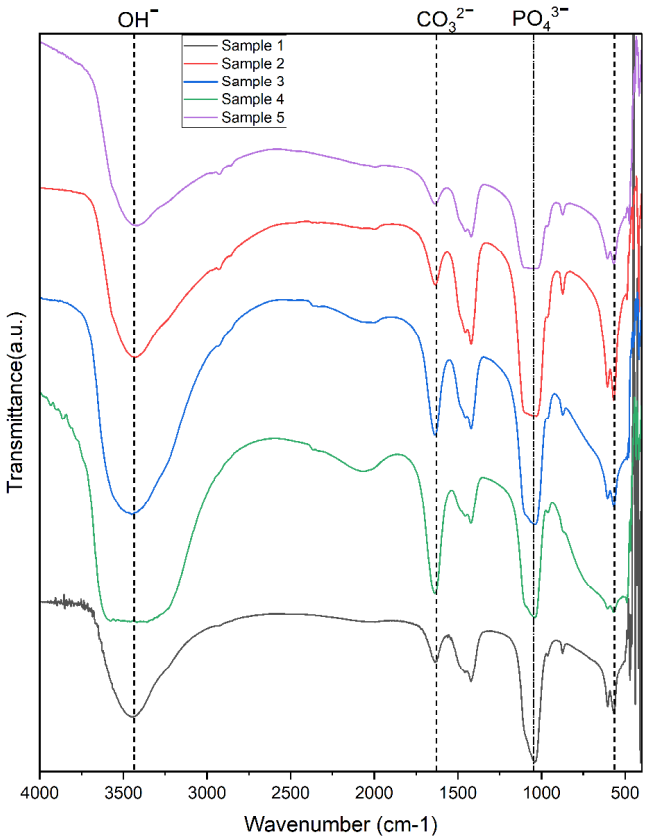

Fig. 4. FT-IR spectrum of five HA samples

The results obtained by "Bruker Alpha II" compact FTIR spectrometer confirm the molecular bond structure in the samples aligns with the characteristic vibrations of hydroxyapatite.

As in Fig. 4. clear IR spectra for HA show vibration at wavenumber 3446 cm^{-1} which corresponds to vibration of OH^-. Absorption bands for the PO_4^{3-} can be observed at around wavenumber 1038 cm^{-1}, 604 cm^{-1} and 567 cm^{-1} for all the HA samples. Bands between 1038 cm^{-1} are due to V3 vibration of PO_4^{3-}, and bands between 567 cm^{-1} are due to n4

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type vibration of PO_4^{3-}. These results confirm the availability of functional groups in synthesized HA samples.[7], [8]

Frequencies observed at around 1600 cm^{-1} -1680 cm^{-1} are due to CO_3^{2-} groups. CO_3^{2-} ions are within the HA samples, that can be due to CO_3^{2-} impurities in $Ca(OH)_2$ and atmospheric carbon dissolution during the synthesis process as it was conducted under ambient environmental conditions.[7], [9]

C. XRD Analysis

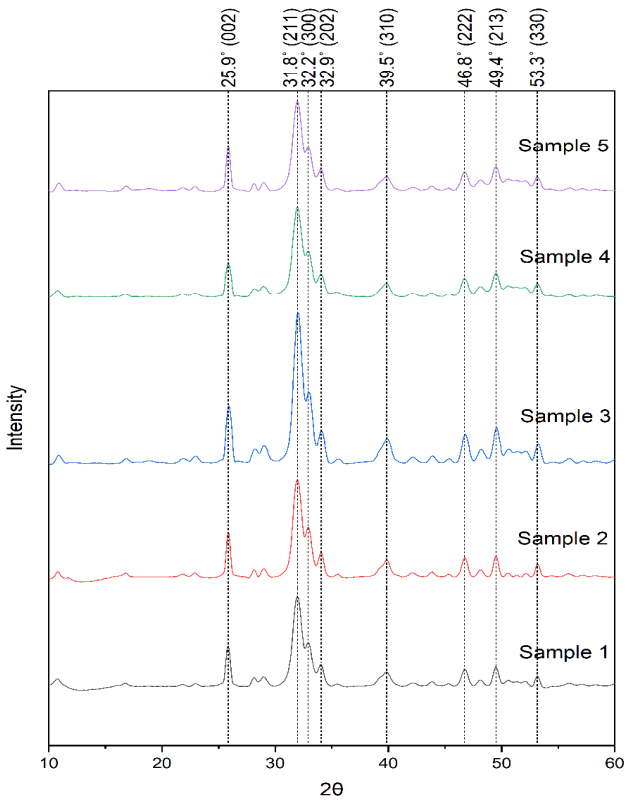

Fig. 5. XRD spectrum of five HA samples

Fig. 5. displays the X-ray diffraction patterns for powdered nano HA acquired in 2θ angle ranging 10˚ to 60˚ from the "Bruker D8 Advance" for all 5 synthesized nano HA samples. The X-Ray diffraction patterns exhibited by the synthesized nano HA samples demonstrate a strong resemblance to the X-ray diffraction patterns for Hydroxyapatite. Corresponding indices (*hkl)* for the peaks are marked in the graphs. The average crystallite size for each sample was calculated using peak intensities, corresponding 2θ angle, wavelength, and the Scherrer's equation.[4]

Each XRD pattern complies with the characteristic peaks and pattern of XRD for nano HA. Major peaks are observed at 31.8˚, 32.2˚, 32.9˚ which are Major peaks for standard HA. *hkl* values for those peaks are (211), (300), (202) respectively.[10] In addition, other prominent peaks observed strongly comply with the standard XRD peaks of HA. Major peaks and their respective *hkl* value are 25.9˚ (002), 39.5˚ (310), 46.8˚ (222), 49.4˚ (213), 53.3˚ (330).[11]

Table 2. Crystal sizes of five samples

Sample No	1	2	3	4	5
Crystal Size(nm)	8.19	9.45	10.81	11.69	13.74

The calculated average crystallite size of the samples varies as shown in Table 2. It reveals the change of crystal size with supersaturation. Recent studies suggest that higher supersaturation prefers smaller crystals.[12]

D. Laser Particle Analyser

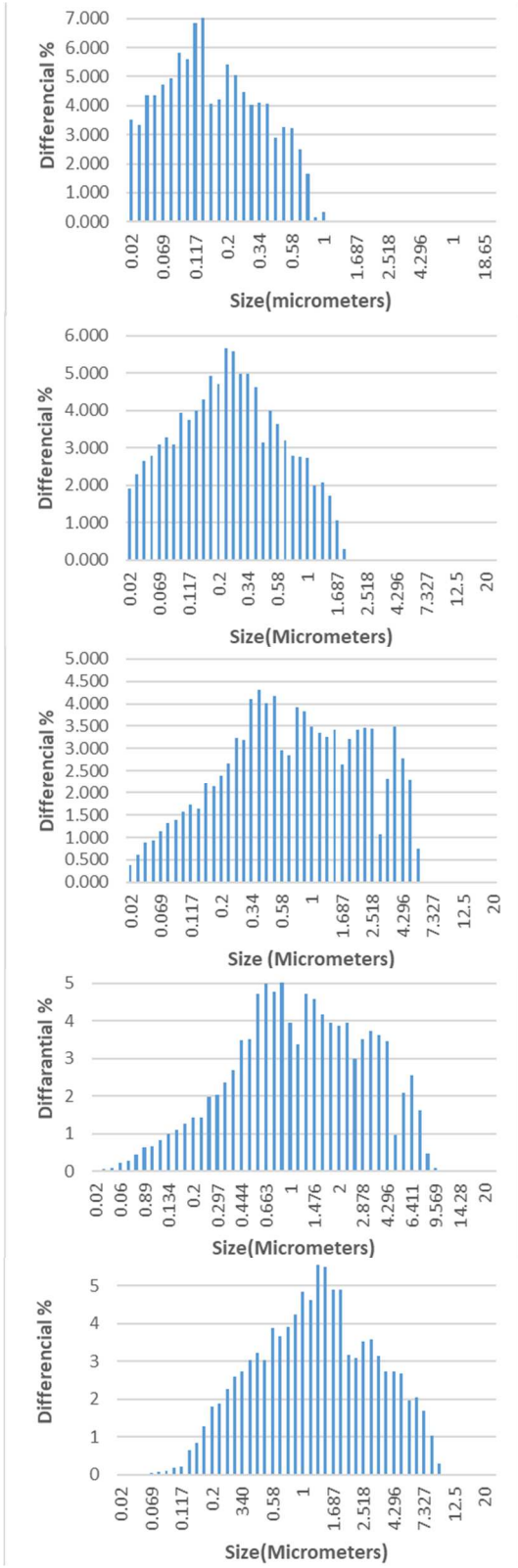

Fig. 6. Particle size distribution of (a) HA sample1, (b) HA sample 2, (c) HA sample 3, (d) HA sample 4 and (e) HA sample 5

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A laser particle analyzer which operate within the range of 20 nanometer – 2,000 micrometres was used to obtain the results in Fig. 6.The instrument provide results according to the method of laser diffraction. Raw data obtained from the instrument was use to express the graphs in Fig. 6. and mean, mode and average particle size for each sample was calculated. Summry of the resust are shwon in Fig. 7.[13]

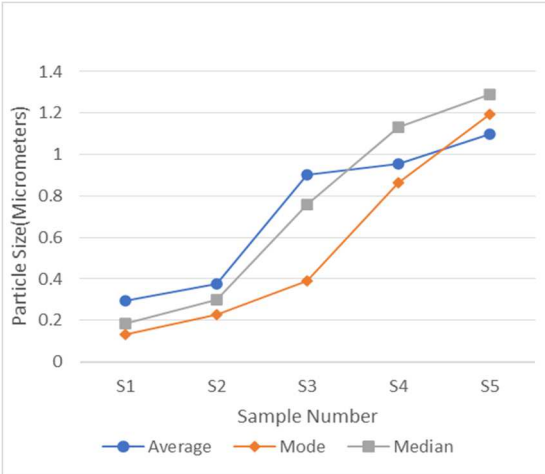

Fig. 7. Variation of average, mode, and median particle sizes for five HA samples.

The results obtained from the Laser particle analyzer have complied with the recent research findings on the supersaturation effect for the final particle size distribution of n-HA. The findings demonstrate the increase in the particle size with the increase of precursor concentration of starting $Ca(OH)_2$ and $H_3(PO_4)_2$. The observed increase in particle size with precursor concentration is due to the decrease in supersaturation as shown in Table 3. According to classical nucleation theory the driving force for the nucleation in a liquid solution is supersaturation. Equation (2) reveals the rate of nucleation strongly depends on supersaturation and it increases with the increase of supersaturation. Results show a decrease in particle size with the increase in supersaturation.

TABLE 3. SUPERSATURATION AND AVERAGE PARTICLE SIZES OF FIVE HA SAMPLES

Sample No	1	2	3	4	5
Ln[s]	70.9858	63.737	52.9636	46.1438	40.4239
Mode Particle size (µm)	0.134	0.228	0.389	0.866	1.192

The lower supersaturations promote a lower amount of nucleation sites in the solution during the nucleation stage. A lower number of nuclei in the medium encourages larger particles by growth.

The fraction of particles is in or close to the micron range in HA Samples 3,4 and 5. Considering the SEM images and XRD results for crystallite size It can be predicted that particles undergo an agglomeration process while ageing. A recent study reveals that Ultrasonic vibration or magnetic stirring does not completely dissolve agglomerations, and it depend on the vibration time. However, ultrasonic vibration and magnetic stirring failed to remove agglomerations completely.[5]

E. EDAX Results

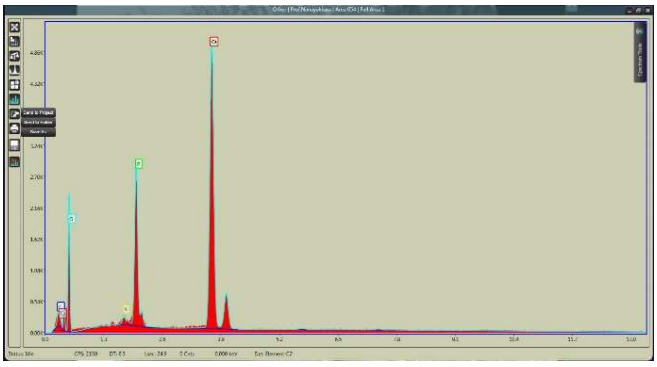

Fig. 8. EDAX spectrum of HA sample 1

EDAX elemental analysis results by the EDAX electron detector in the Scanning Electron Microscope. confirm the elements and their compositions in samples are comply with the Ca/P ratio of Hydroxyapatite.

The Stoichiometry of HA contains a Ca/P ratio of around 1.67.[14], [15] These HA samples were pH neutralized and washed several times which only contains the precipitated solid particles. Then the sample was dried to remove excess water. Therefore, the elemental analysis reveals the Ca/P molar ratio in the precipitated solid particles only.

VI. CONCLUSIONS

Nano-sized HA particles have been synthesized by manipulating the supersaturation of the solutions using the wet chemical synthesis method. It was observed that the least average-sized particles were observed in the highest supersaturated sample.

One significant discovery of the study is the correlation between supersaturation and the average particle size of the precipitated Hydroxyapatite nanoparticles. It was found out that the highest supersaturation, which resulted in the precipitation of smaller-sized nano particles prefers having the smallest crystallite size. According to the classical nucleation theory for precipitation in wet chemical synthesis method, the driving force for nucleation is supersaturation. Where the higher nucleation rate can be observed in higher supersaturations. Higher nucleation rate provides a higher number of nuclei generation in the solution. A greater number of nuclei in the solution lead to a shortage of dissolved ions in solution. That results in smaller sized particle formation by growth. The effect of growth is minimum when the number of nuclei in the sample is higher. Moreover, the least crystallite size was found in the least particles. This is confirmation of having smaller particles. However, the Laser Particle Analyzer results and SEM images provide sufficient details about agglomeration of final particles. That results in some fractions of particles are in micron range. However ultrasonic vibration does not completely dissolve those agglomerations.

These results emphasize the influence of supersaturation on the particle size and the crystal size of Hydroxyapatite nanoparticles synthesized using wet chemical precipitation method. By varying the supersaturation during the wet

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chemical synthesis, it becomes possible to control the resulting crystal size and nano particle size, which offers the opportunity to tailor characteristics and properties of Hydroxyapatite nano particles for a specific application.

VII. Acknowledgment

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